



Batterjee Medical College
For Science and Technology

An approach to a case of cough Lung Abscess, Bronchiectasis & Cystic Fibrosis

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- Definition
- Etiology
- Pathophysiology
- Clinical Features
- Diagnostics
- Management

Learning Objectives

Knowledge:

- Define bronchiectasis, lung abscess and cystic fibrosis.
- Explain the pathophysiology and vicious cycle of bronchiectasis, lung abscess and CFTR dysfunction.
- List the congenital, infectious, obstructive, and systemic etiologies of bronchiectasis, lung abscess and CFTR.
- Identify key clinical features and multiorgan complications, including respiratory insufficiency, digital clubbing, exocrine pancreatic insufficiency, CF-related liver disease, and infertility.
- Summarize management approaches, including airway clearance techniques, CFTR modulators, targeted antibiotic regimens (particularly against *Pseudomonas aeruginosa*), and replacement therapies for bronchiectasis, CFTR and lung abscess.

Skills

- Interpret diagnostic imaging, specifically identifying key radiologic signs on chest X-ray and HRCT for lung abscess, bronchiectasis and CFTR..
- Evaluate clinical cases presenting with chronic cough, dyspnea, and hemoptysis to form an accurate differential diagnosis.
- Select and interpret initial diagnostic tests appropriately.
- Formulate comprehensive evidence-based treatment plans tailored to acute exacerbations versus chronic maintenance.

Values:

- Demonstrate a professional attitude by adhering to institutional training rules and clinical guidelines.
- Appreciate the burden of chronic illness on patients and families, emphasizing long-term supportive care, nutrition optimization, and preventive immunization.

Suppurative lung Disease

Suppurative : Means Pus formation

Suppurative lung disease: a disease characterized by chronic inflammation and the formation or discharge of pus (suppuration) within the airways or lung tissue.

Main Types:

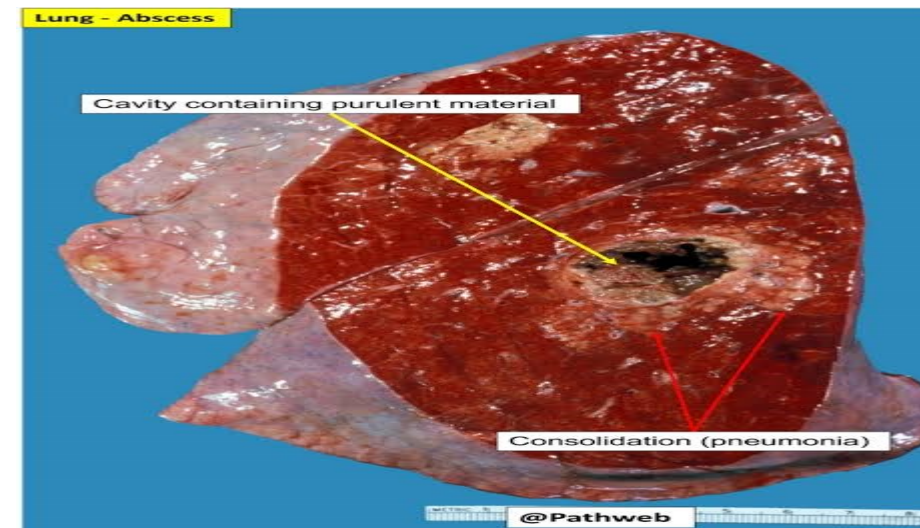
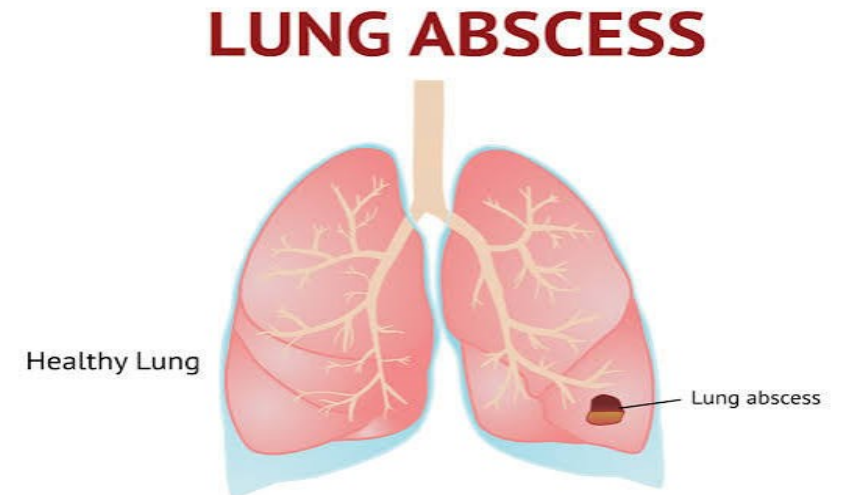
1. Lung Abscess
2. Bronchiectasis
3. Cystic fibrosis
4. Pneumonia

Lung Abscess

Definition: a localized collection of pus and necrotic tissue within lung parenchyma caused by microbial infection.

Primary lung abscess: abscess in normal lung parenchyma, typically due to aspiration

Secondary lung abscess: abscess in a patient with immunocompromise or preexisting disease (e.g., lung neoplasm), or due to hematogenous spread (e.g., septic emboli, IV drug use)



Lung Abscess

Causative organism:

Bacterial (pyogenic lung abscess)

- Most commonly caused by anaerobic bacteria that colonize the oral cavity.
(e.g., *Peptostreptococcus* spp., *Prevotella* spp., *Bacteroides* spp., *Fusobacterium* spp.)
- Less commonly caused by aerobic bacteria, such as:
 - ☐ Microaerophilic streptococci and viridans streptococci.
 - ☐ *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, group A *Streptococcus*, *Streptococcus anginosus*, or *Haemophilus influenzae*
 - ☐ Parasitic: e.g., *Entamoeba histolytica*
 - ☐ Fungal: e.g., *Aspergillus* spp., *Cryptococcus* spp., *Coccidioides* spp., and *Histoplasma* spp.

Lung Abscess

Risk factors for lung abscess

- Risk factors for aspiration:
 - Impaired consciousness
 - Impaired swallow in neurological disorders and vocal cord paralysis
- Increased oropharyngeal bacterial growth (e.g., periodontal disease, dental abscesses, tonsillitis)
- Bronchial obstruction (e.g., lung cancer, foreign body aspiration, bronchial stenosis)
- Immunocompromised state
- Pneumonia, bronchiectasis
- Impaired respiratory mucus clearance (e.g., cystic fibrosis)

Pathophysiology

- PATHOLOGY AND PATHOGENESIS:

Aspiration of infected material or foreign body



Pneumonitis impairs drainage of fluid or aspirated material



Inflammatory vascular obstruction



Tissue necrosis, liquefaction



Abscess formation

Clinical Features

Onset

symptoms can evolve over weeks to months, but may also be acute in onset.

- Acute: symptoms present for < 4–6 weeks
- Chronic: symptoms present for > 6 weeks

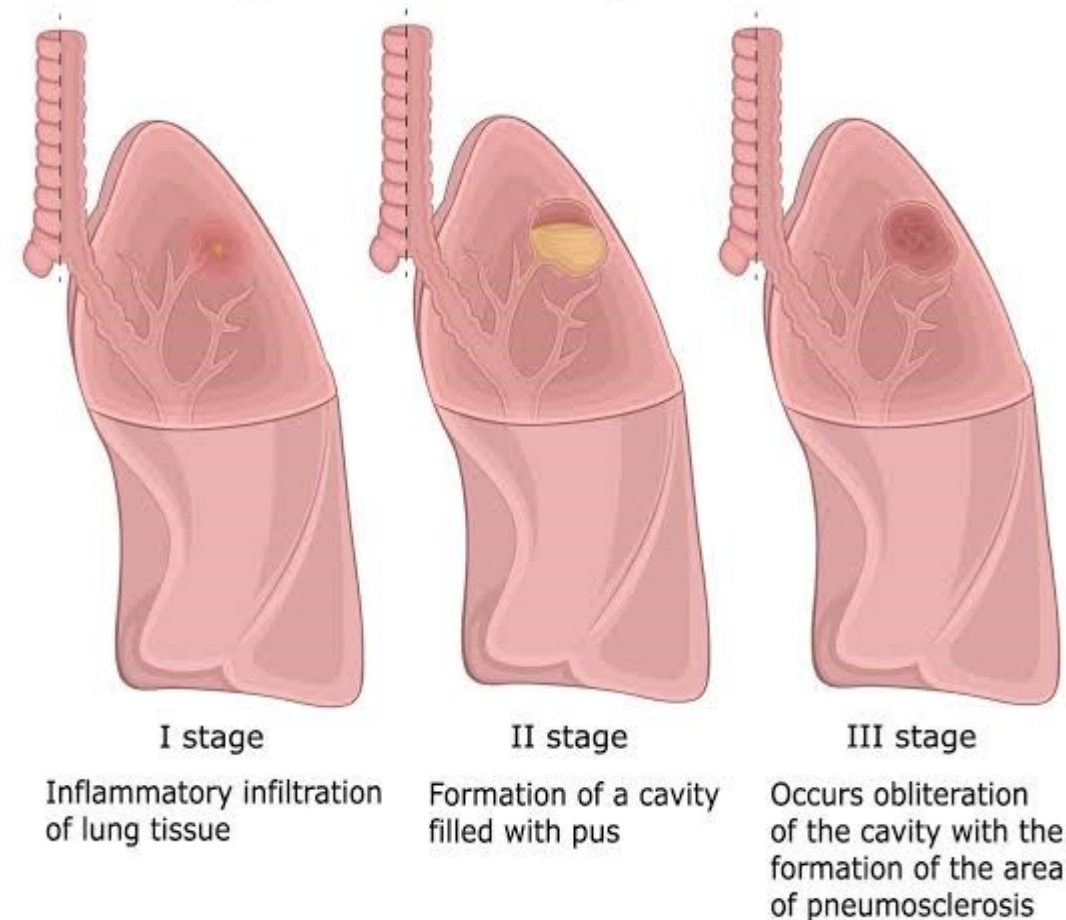
Characteristic symptoms

- Fever
- Cough with production of foul-smelling purulent sputum (increase on lying on healthy side)
- Night sweats
- Hemoptysis
- Anorexia, weight loss, fatigue
- Pleuritic chest pain

Pulmonary examination findings

- Digital clubbing in chronic abscess
- Dullness to percussion over the affected area
- Bronchial breath sounds may be present over the lung abscess

Stages of Lung Abscess



Diagnosis

CBC: elevated WBC

CRP & procalcitonin: Elevated

Blood culture

Frequently negative in anaerobic infections

Gram stain and culture

Of sputum (or bronchoalveolar lavage)

Do not wait for culture. Start empirical Antibiotics immediately after obtaining samples for culture, and adjust once culture results are available.

HIV screening: if Atypical pathogens are cultured

Diagnostics for the underlying cause

Bronchoscopy for suspected malignancy or lower airway foreign body

Echocardiography for suspected endocarditis

Radiology

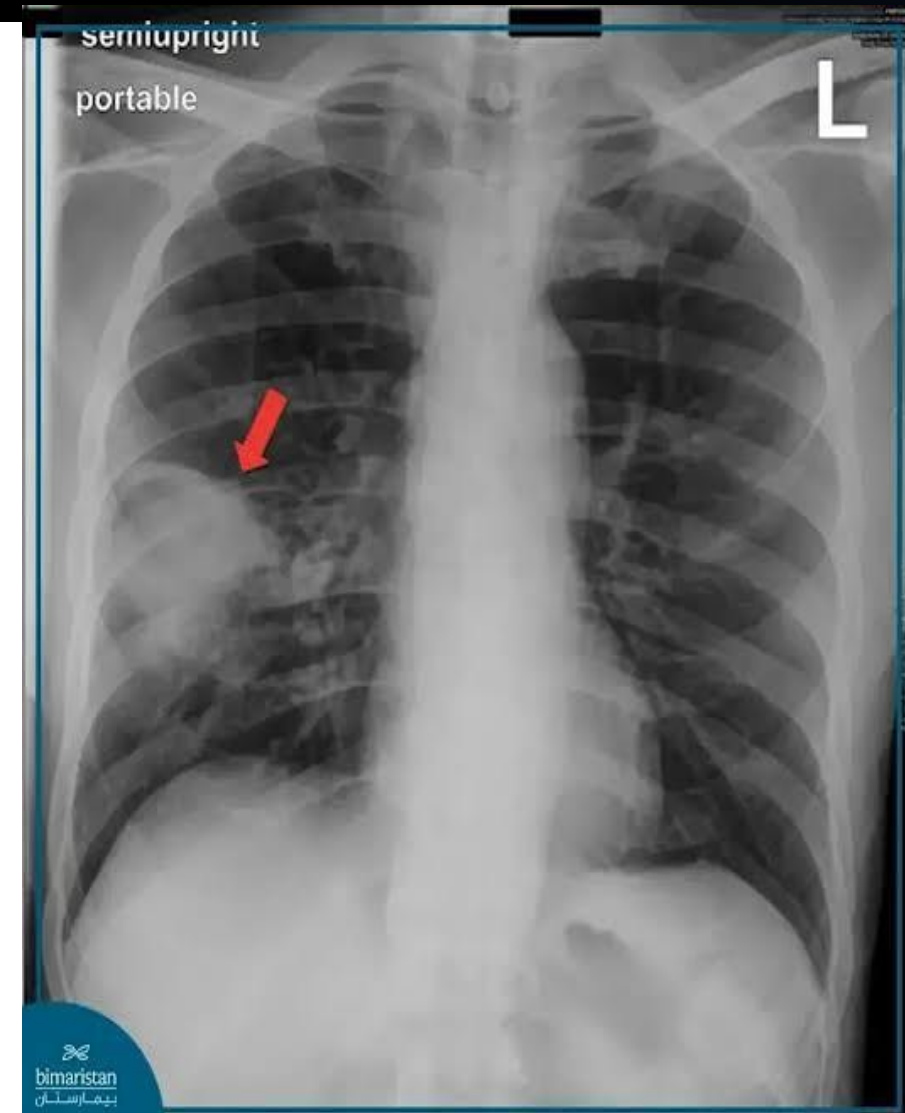
Chest x-ray:

CT chest with IV contrast: to confirm the diagnosis Findings

Common findings on x-ray and CT

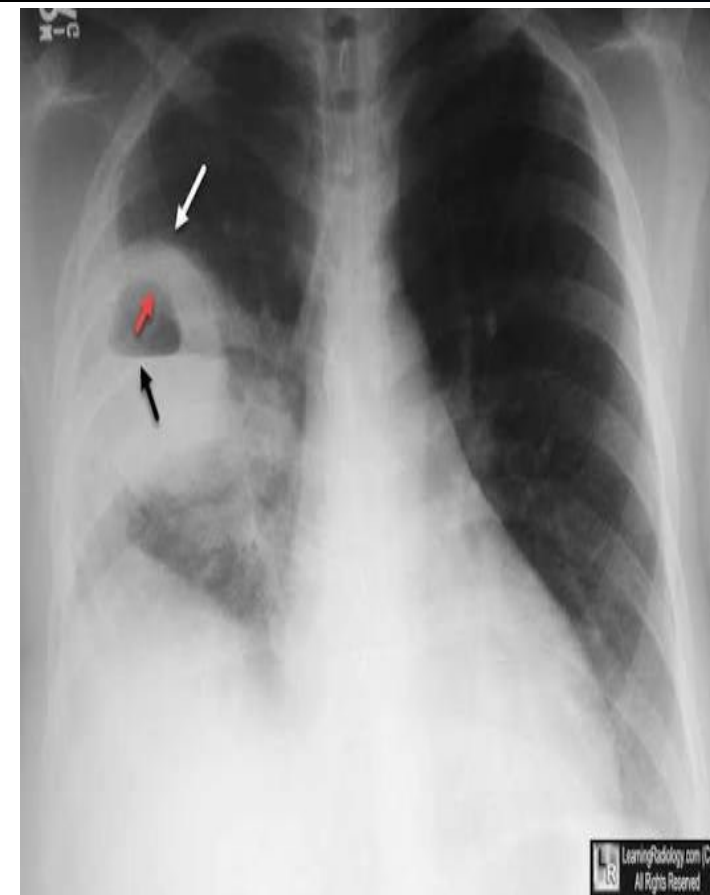
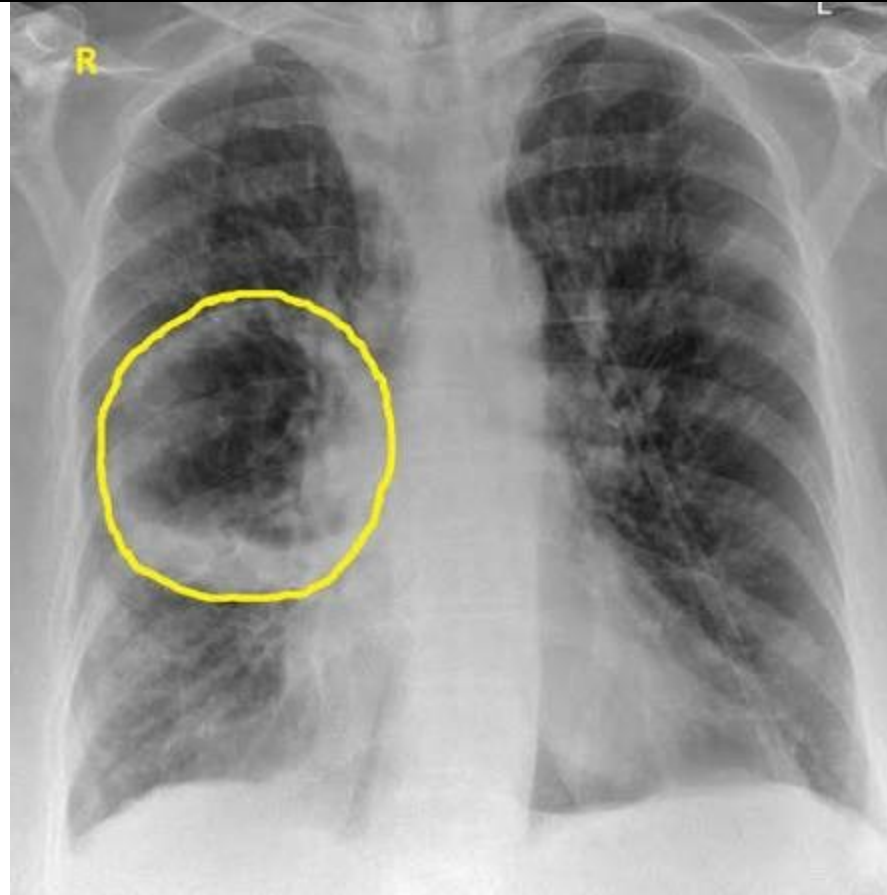
Spherical intraparenchymal cavity with thick irregular walls

Air-fluid level within the cavity



Location of abscess

- Due to aspiration: typically unilateral Right middle lobe (typically caused by aspiration in prone)
- Posterior segments of the upper lobes or the superior segments of the lower lobes (typically caused by aspiration in the recumbent position).
- Due to hematogenous dissemination: typically bilateral and multiple



Treatment

Antibiotic therapy:

- Start Broad-spectrum antibiotics with anaerobic coverage are recommended (e.g., ampicillin-sulbactam, carbapenems, or clindamycin).
- Ensure antibiotic cover gram-positive cocci in lung abscesses that are likely due to IV drug use.
- Adjust antibiotic therapy as needed based on culture results and clinical response.
- Antibiotics are typically continued for 3–6 weeks or until clinical and radiological improvement is seen.

Interventional therapy

•Indications

- Large abscess
- Significant hemoptysis
- Inadequate response despite appropriate antibiotic therapy, characterized by:
 - Persisting fever and/or purulent sputum
 - Inadequate signs of resolution on imaging

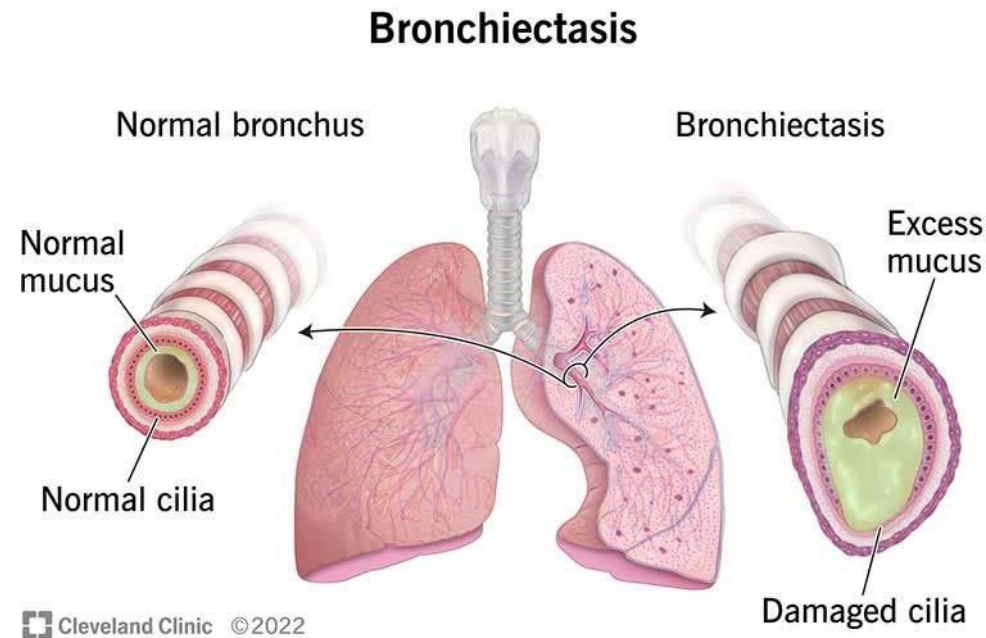
•Options

- First line: bronchoscopic drainage or image-guided percutaneous drainage
- Rarely : surgical resection (segmentectomy, lobectomy, pneumonectomy)

Bronchiectasis

Definition:

There is permanent, abnormal dilation and destruction of bronchial walls with chronic infection, sputum production, airway collapse, and ciliary loss/dysfunction leading to impaired clearance of secretions.



Bronchiectasis

Bronchiectasis requires the combination of two important processes taking place in the bronchi: either **local infection or inflammation** along with either **inadequate clearance of secretions, airway obstruction, or impaired host defenses**. These processes result in the permanent dilation of airways.

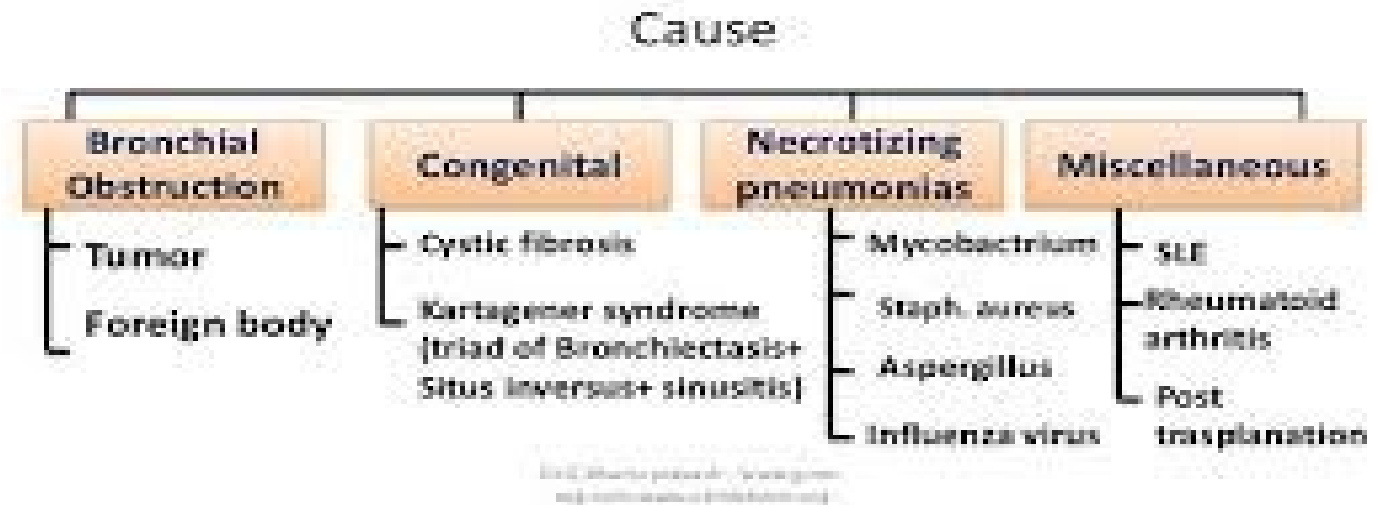
Bronchiectasis

Etiology:

- Recurrent infection
- Autoimmune disease
- Humoral immunodeficiency
- Disorders of secretion clearance or mucous plugging
 - **Cystic fibrosis**
 - Primary ciliary dyskinesia
 - Allergic bronchopulmonary aspergillosis (ABPA)
 - Kartagener syndrome
 - Smoking: associated with poor ciliary motility

4. Bronchiectasis

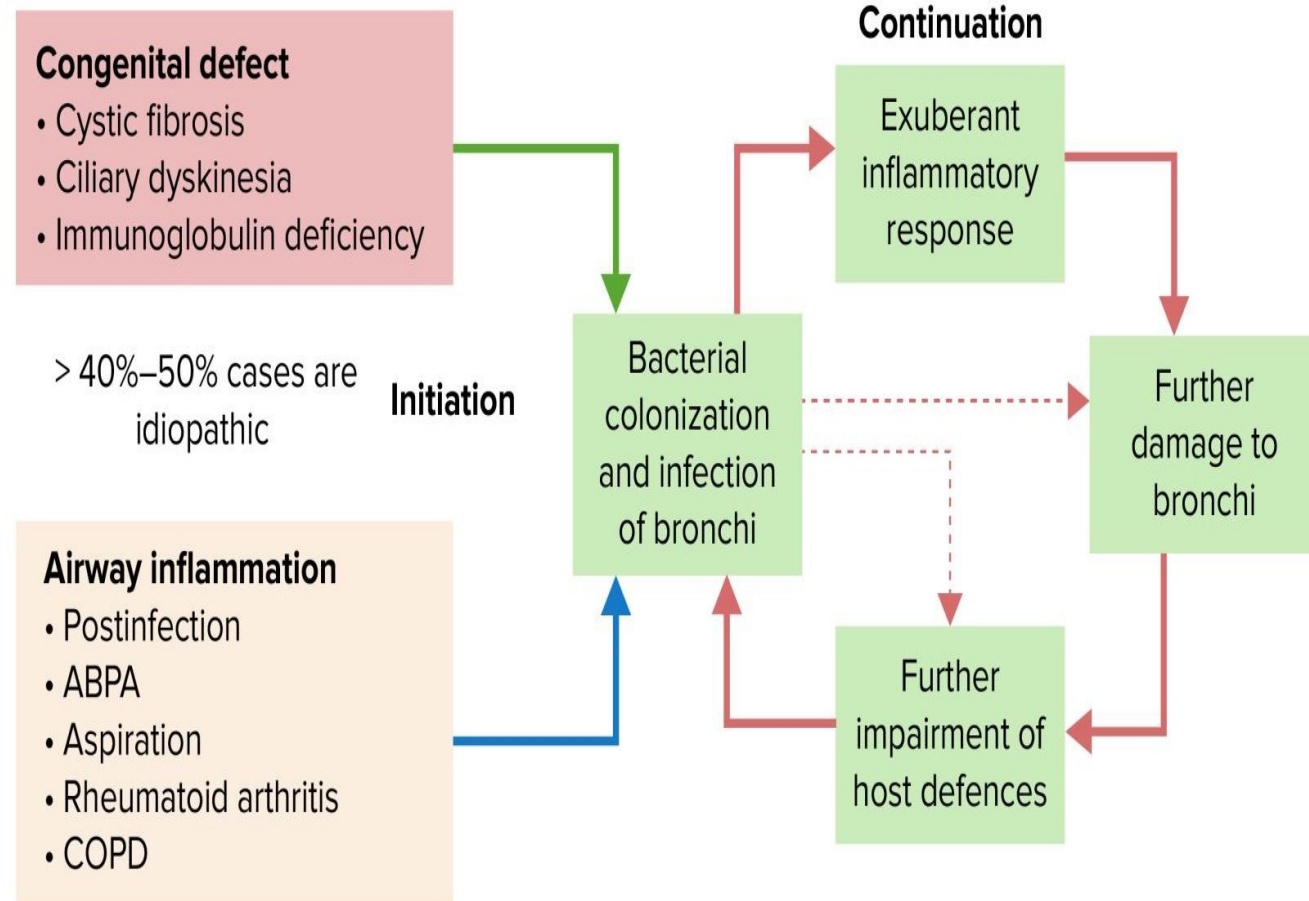
- Abnormal permanent airways dilation resulting from chronic necrotizing infection is called bronchiectasis.



Bronchiectasis

Etiology:

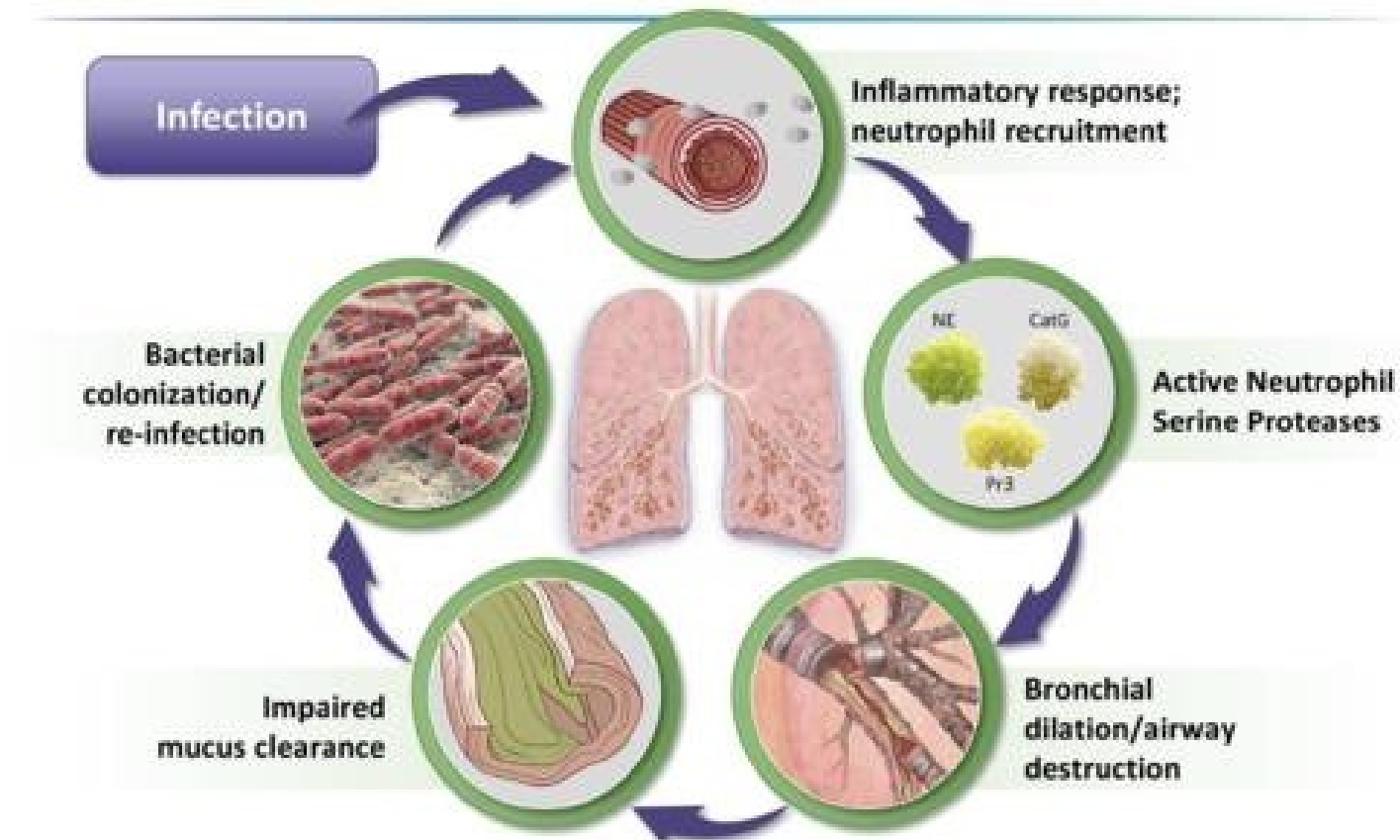
- Bronchial narrowing or other forms of obstruction
 - COPD
 - Aspiration
 - Tumors
 - α 1-antitrypsin deficiency
 - Other congenital and acquired conditions (e.g., tracheomalacia, bronchogenic cyst)
- Immunodeficiency (e.g., hypogammaglobulinemia, HIV)
- Chronic inflammatory diseases (e.g., rheumatoid arthritis, Sjogren syndrome, Crohn disease)



Bronchiectasis

Pathophysiology:

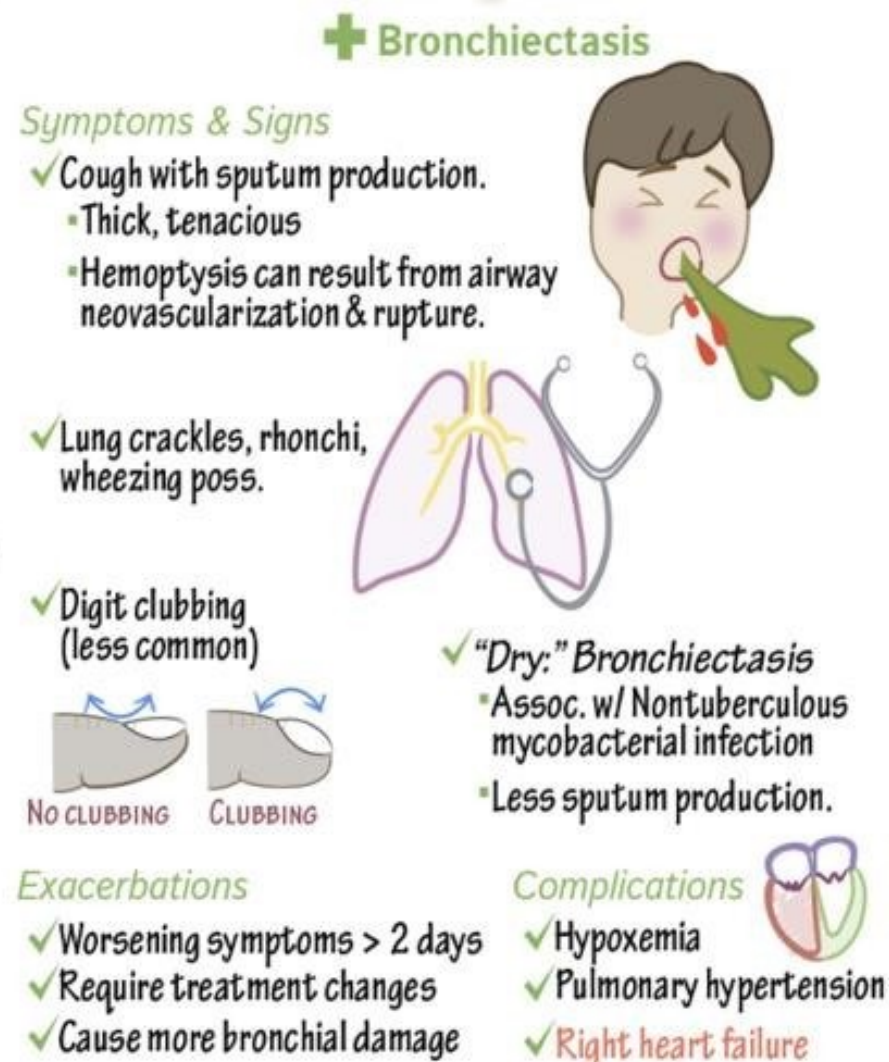
Bronchiectasis is a vicious cycle of infection and inflammation



Bronchiectasis

Clinical Features:

- Chronic Cough (large amount of mucopurulent sputum)
- Dyspnea
- Hemoptysis mild & self limiting
- Nonspecific symptoms (i.e., fatigue, weight loss, pallor due to anemia)
- **Clubbing** of nails (uncommon)
- **Exacerbations of bronchiectasis**
 - Recurrent bouts of pneumonia and acute bacterial infection of sections of dilated bronchi
 - Frequently due to *Pseudomonas aeruginosa*
 - Features
 - Increased production of mucous above baseline
 - Low-grade fever



Bronchiectasis

Clinical Features:

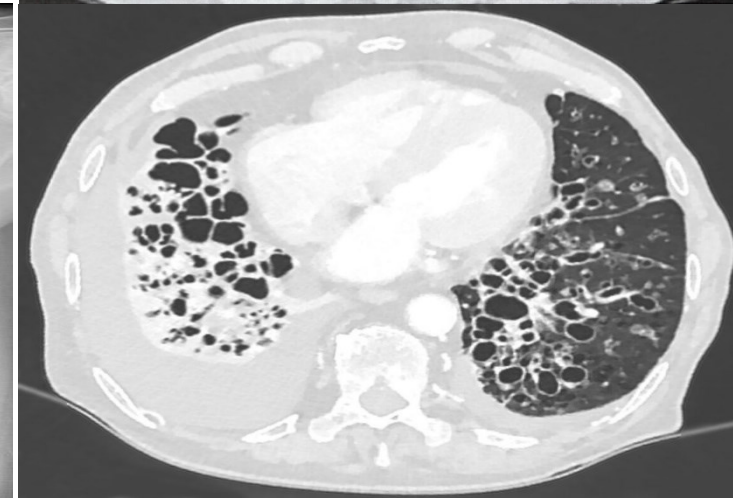
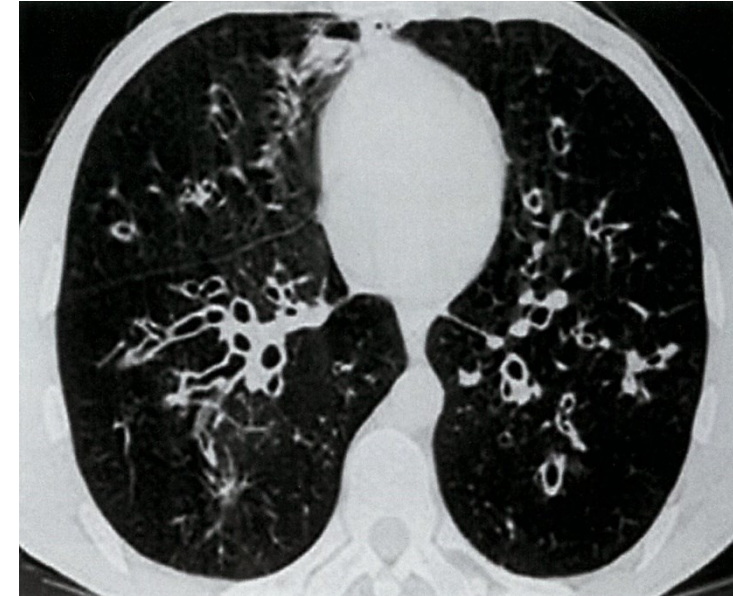
Auscultation

- Crackles and rhonchi
- Wheezing (due to obstruction from secretions, airway collapsibility, or a concomitant condition)

Bronchiectasis

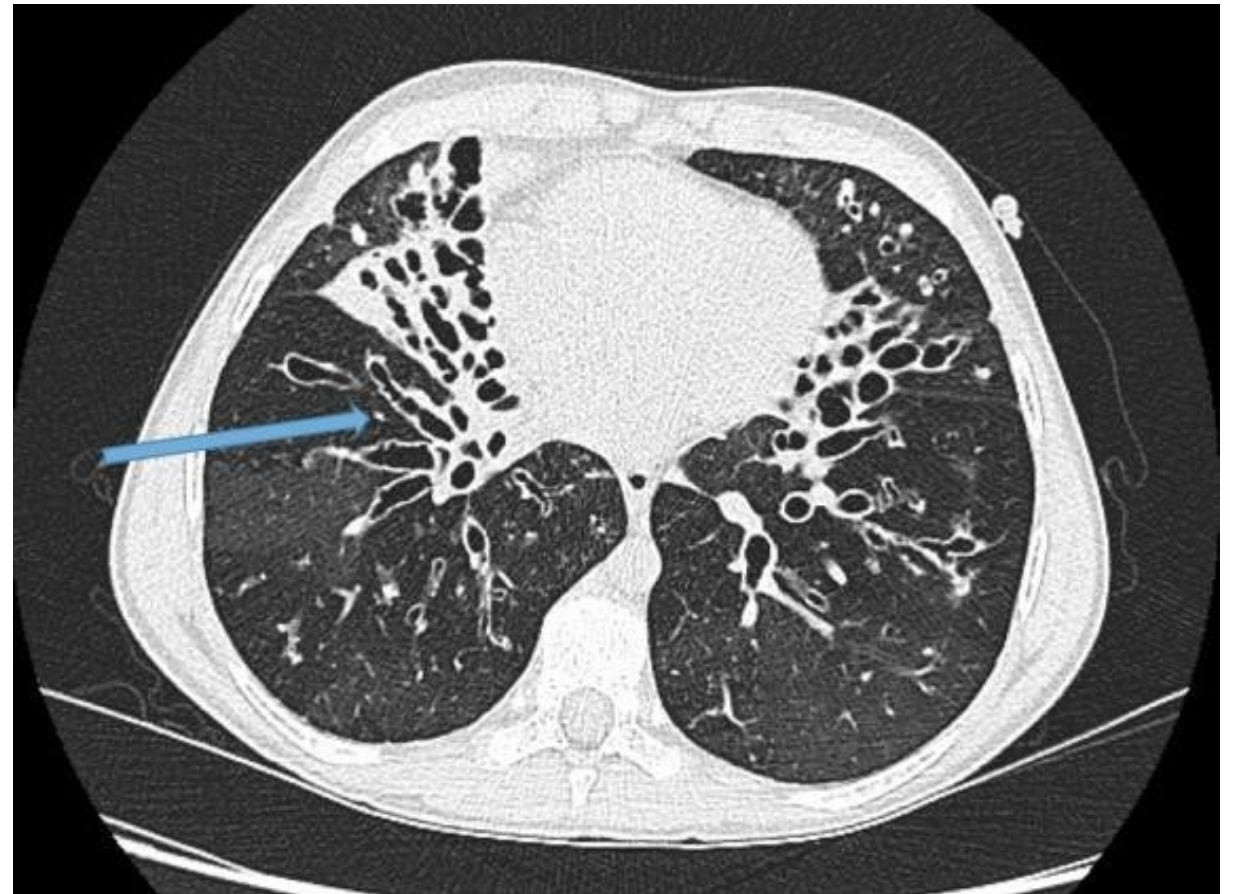
Diagnostics:

- High Resolution CT (TRAM sign and signet ring sign)
- CXR abnormal but not specific may not show mild cases
- PFTs (obstructive pattern)
- Sputum cultures



TRAM sign

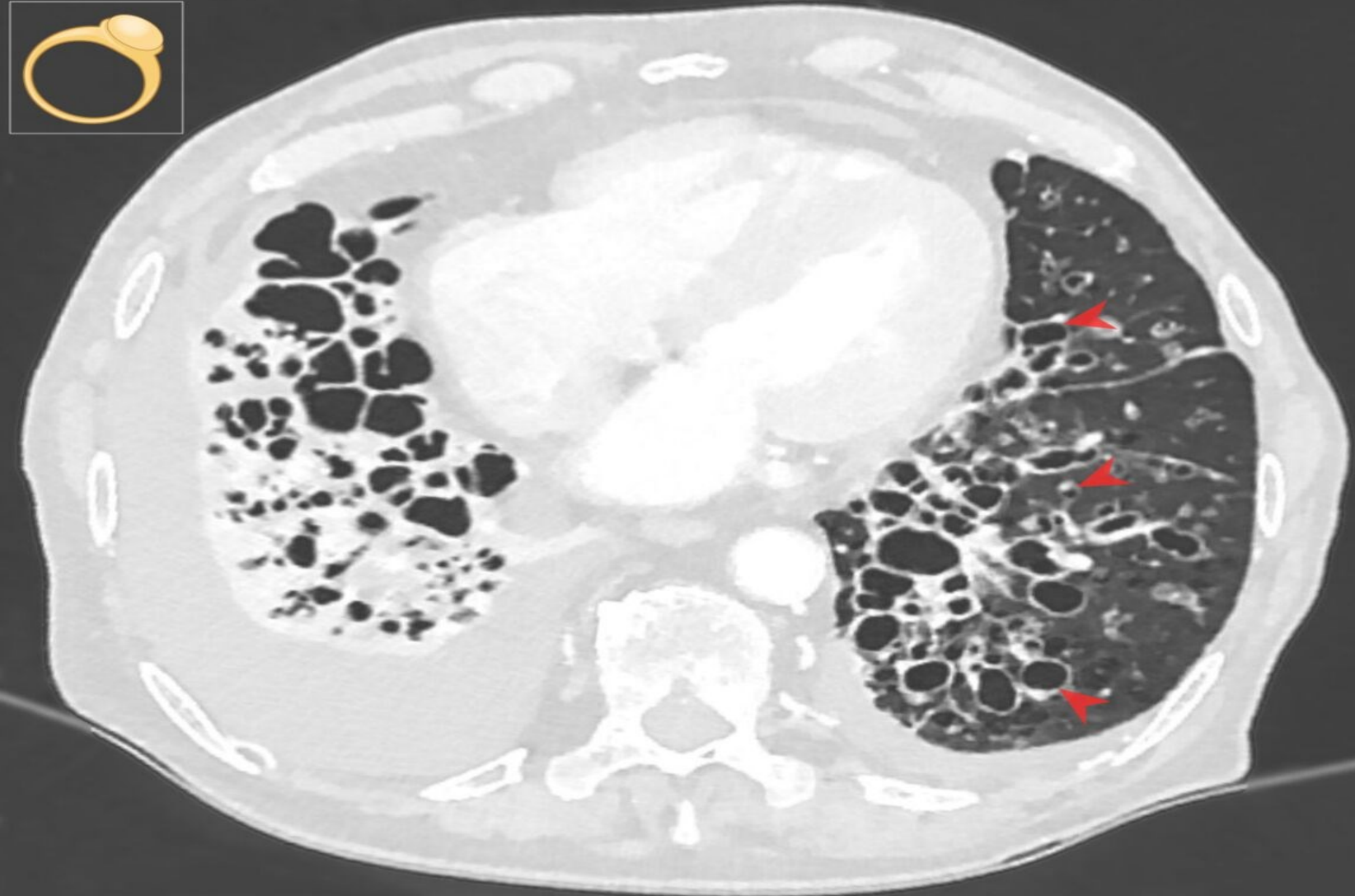
Numerous parallel lines representing thickened bronchial walls demarcate lucent air within dilated bronchi.



Signet Ring sign



Multiple dilated bronchi in the lower lobes and lingula are accompanied by inflammatory wall thickening. Many bronchi are larger than the adjacent pulmonary arteries



Bronchiectasis

Identification of the underlying etiology

In many cases, it is not possible to reach a specific diagnosis and the disease may be classified as idiopathic.

•Laboratory studies

- CBC: Findings are variable and depend on the underlying etiology.
 - ↓ Hb, Variable leukocyte levels, ↑ Thrombocytes
- ABG: may show hypercapnia
- Quantitative measurement of serum immunoglobulins and electrophoresis to exclude immunodeficiencies.
- *Aspergillus fumigatus* IgG and IgE: to exclude ABPA
- Serum AAT level: to evaluate for AAT deficiency

•Sputum culture and smear: culture induced sputum or bronchoalveolar lavage

•Spirometry: not needed to establish a diagnosis but useful to monitor disease progression

- Mixed obstructive/restrictive pattern
- Restrictive pattern (i.e., ↓ FEV₁ and FVC with normal or ↑ FEV₁/FVC ratio)
- Obstructive pattern (i.e., ↓ FEV₁/FVC ratio)

Bronchiectasis

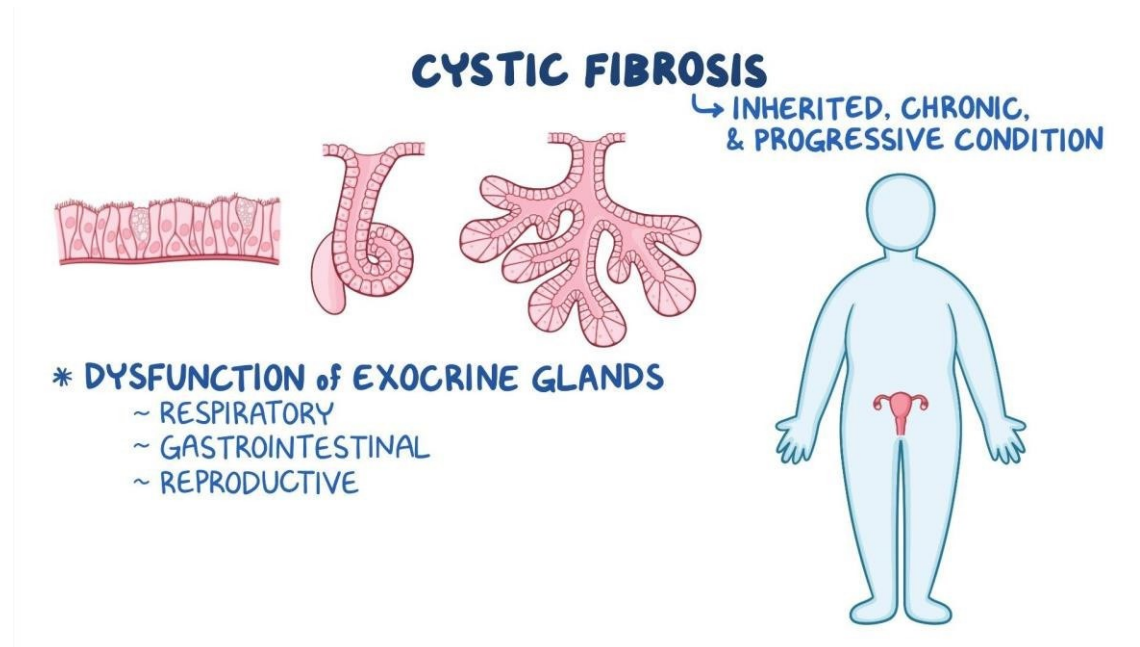
Management:

- **Antibiotics:** to suppress bacterial growth and to reduce symptoms and exacerbations as a measure of secondary prevention in patients with ≥ 3 exacerbations per year. Antibiotic therapy should be administered for at least 3 months and may be extended based on clinical response
- Immunization (influenza and pneumococcal pneumonia)
- Chest Physiotherapy
- Mucolytics
- Bronchodilator therapy is of limited benefit (considered case by case)
- Oxygen therapy
- Surgical resection: (lobectomy in cases of bronchial Hge.)

Cystic Fibrosis

Definition:

- Defect in chloride channel protein causes impaired chloride and water transport, which leads to excessively thick, viscous secretions in the respiratory tract, exocrine pancreas, sweat glands, intestines, and genitourinary tract

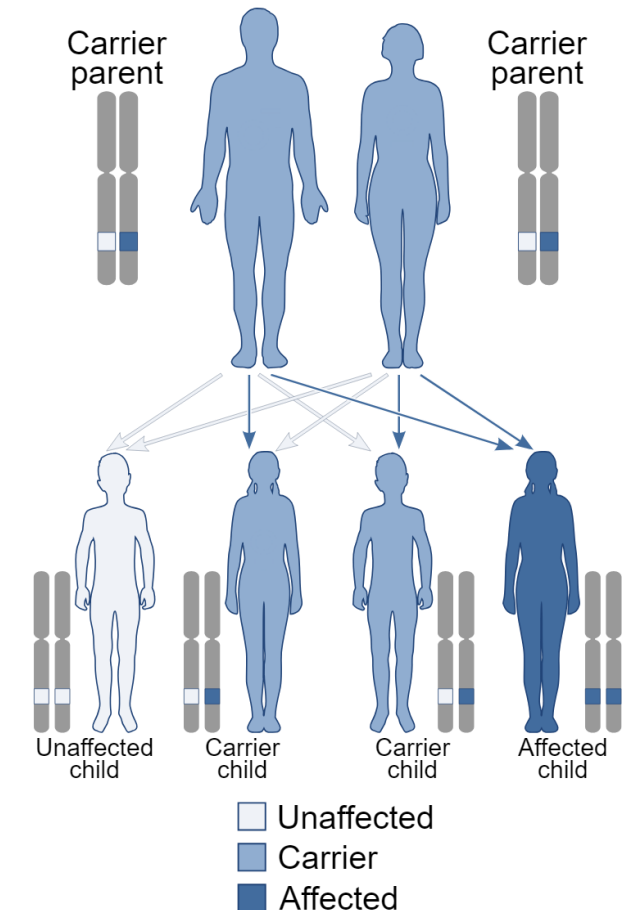


Cystic Fibrosis

Etiology:

- CF is a hereditary autosomal recessive disorder
- Caused by defective CFTR (cystic fibrosis transmembrane conductance regulator) protein caused by mutations in the *CFTR* gene
- Codon deletion that leads to the absence of phenylalanine (F) in position 508 in CFTR protein.

Autosomal recessive



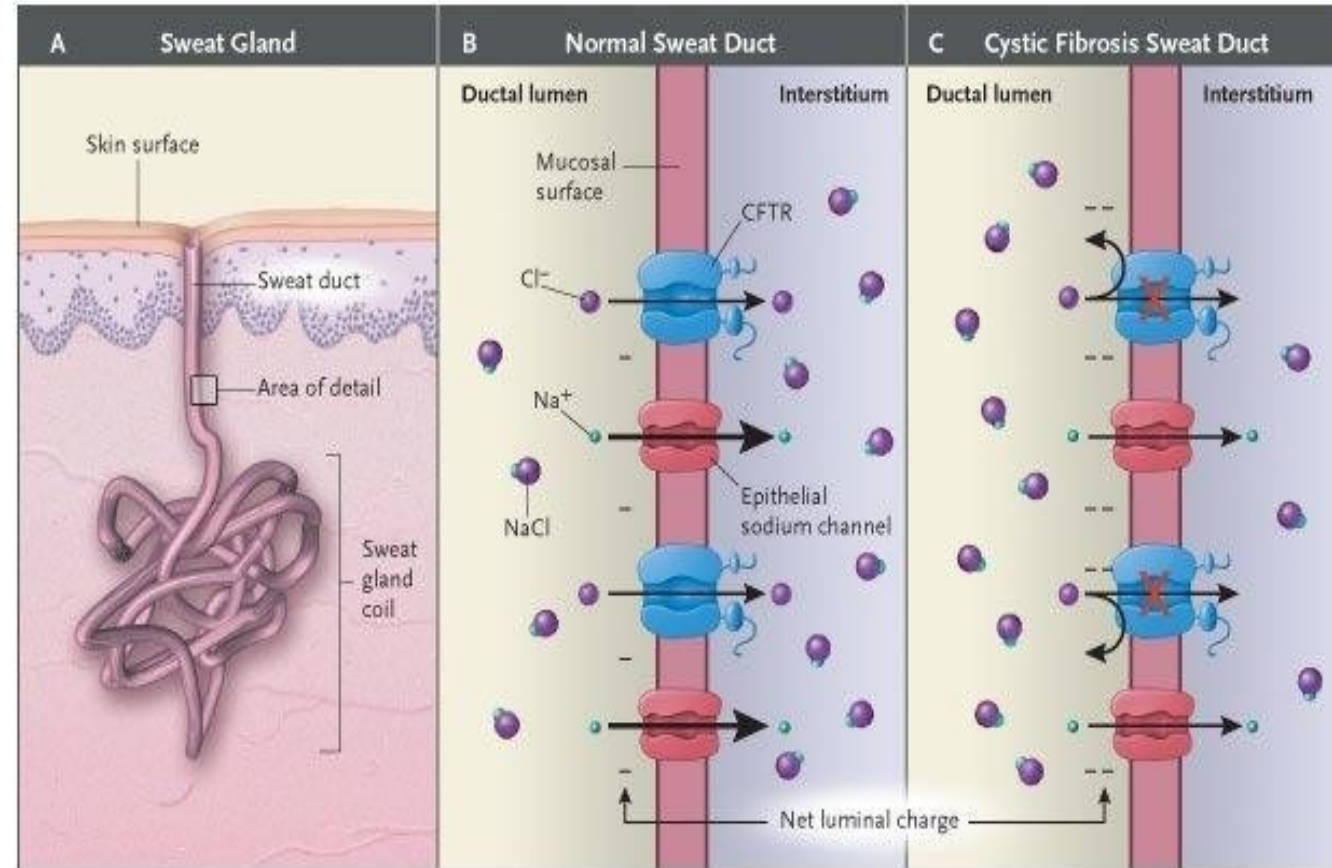
Cystic Fibrosis

Pathophysiology:

Mutated *CFTR* gene → misfolded protein → absence of ATP gated chloride channel on the cell surface of epithelial cells throughout the body.

•In sweat glands

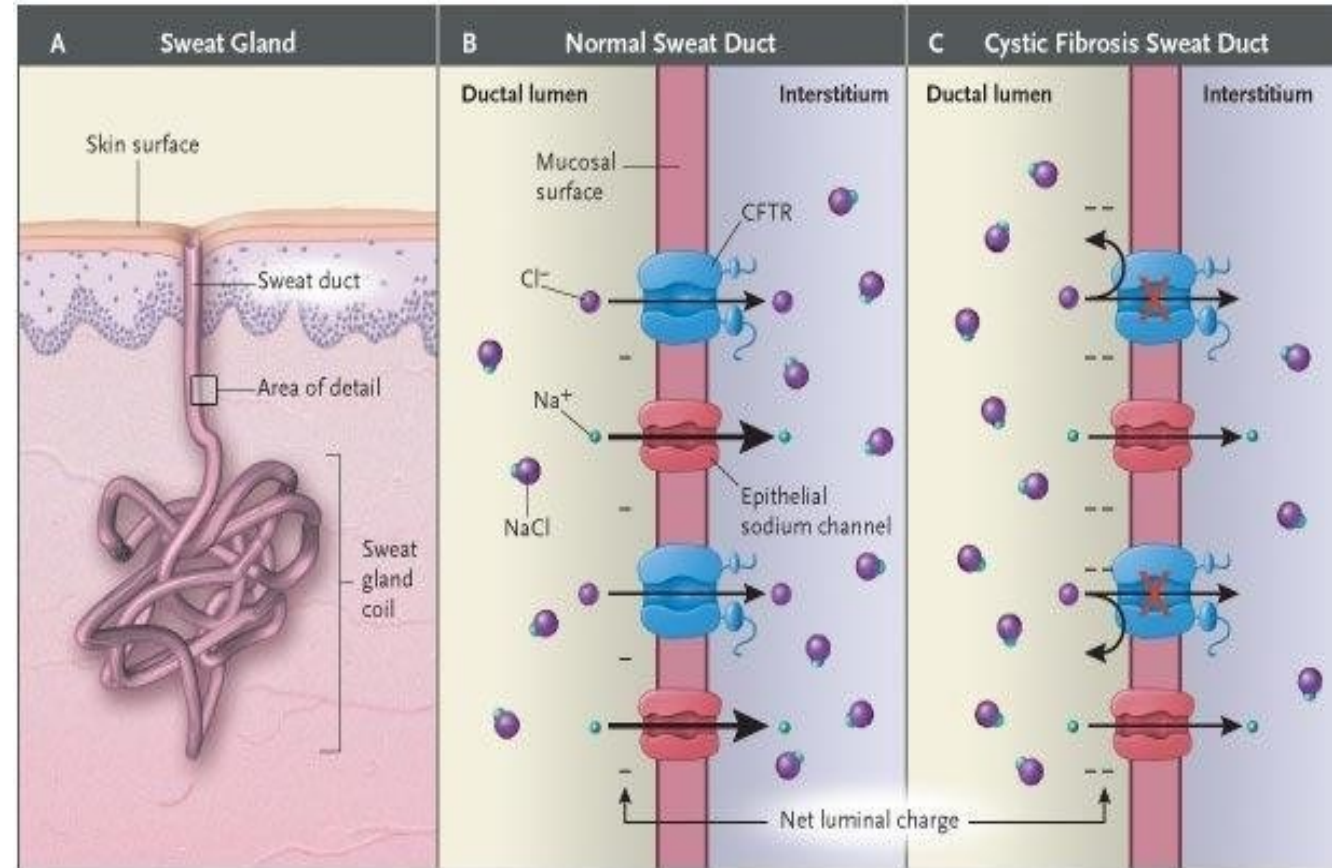
- The chloride channel is responsible for transporting Cl^- from the lumen into the cell (reabsorption).
- **Inability to reabsorb Cl^-** from the lumen of the sweat glands → reduced reabsorption of Na^+ and H_2O → excessive loss of salt and **elevated levels of NaCl in sweat**



Cystic Fibrosis

Pathophysiology:

- In all other exocrine glands (e.g., in the GI tract or lungs) The chloride channel is responsible for transporting Cl^- from the cell into the lumen (secretion).
- Inability to transport intracellular Cl^- across the cell membrane $\rightarrow$ **reduced secretion of Cl^- and H_2O** $\rightarrow$ accumulation of intracellular Cl^- $\rightarrow$ $\uparrow$ Na^+ reabsorption $\rightarrow$ $\uparrow$ H_2O reabsorption $\rightarrow$ formation of **hyperviscous mucus** $\rightarrow$ accumulation of secretions and **blockage of small passages** of affected organs

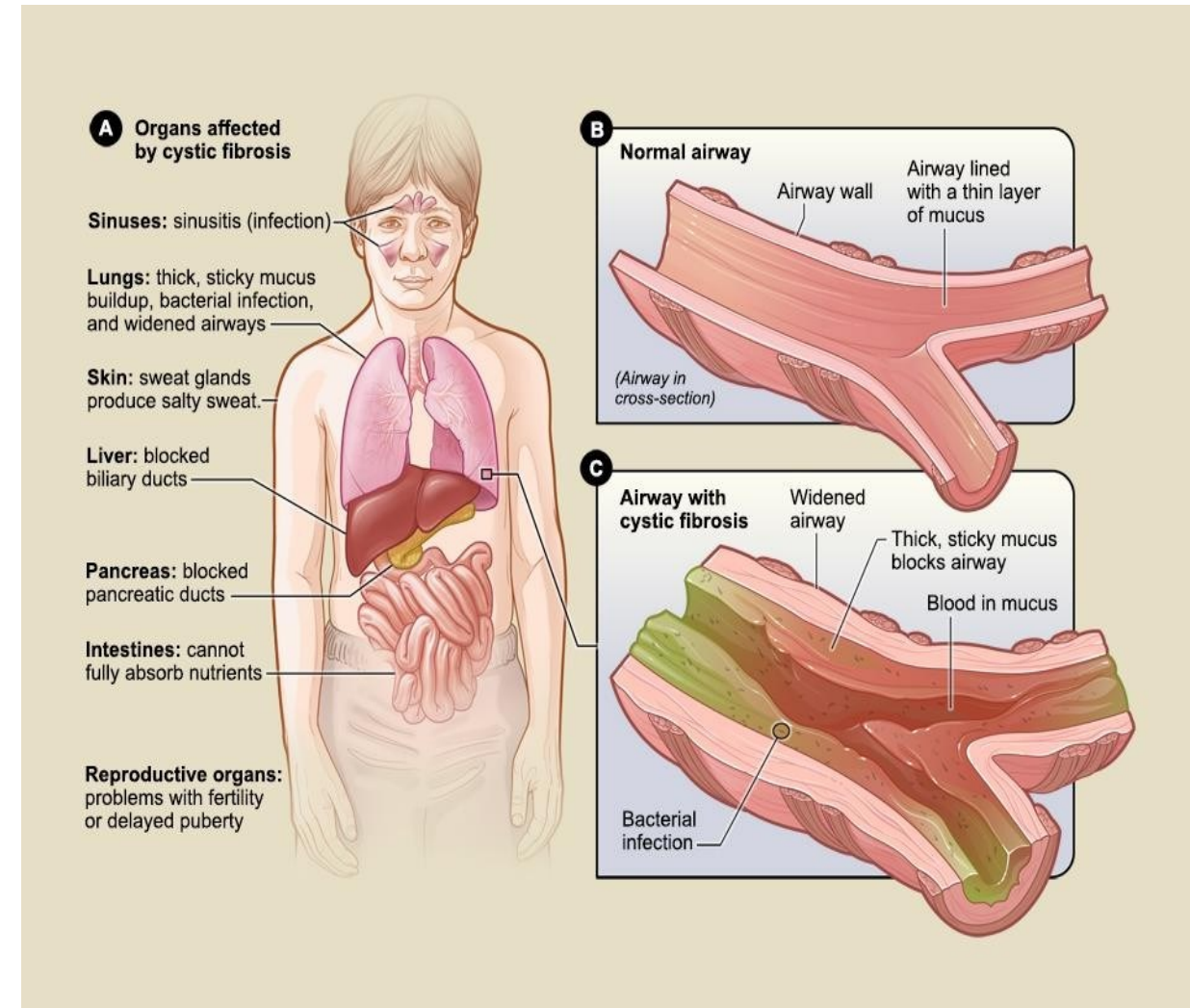


Cystic Fibrosis

Clinical Features:

Respiratory symptoms are common in adulthood.

- **Chronic obstructive lung disease** with bronchiectasis
- **Chronic sinusitis:** nasal polyps may eventually develop
- **Recurrent or chronic productive cough and pulmonary infections.**
 - Increased susceptibility of individuals with CF to opportunistic, potentially life-threatening pathogens (e.g., *Pseudomonas aeruginosa*, *Aspergillus*)
- Pulmonary obstruction and airway hyperreactivity may manifest with expiratory wheezing and/or dyspnea.
- Barrel chest, moist rales, hyperresonance to percussion
- Hemoptysis
- Signs of chronic respiratory insufficiency: digital clubbing associated with chronic hypoxia

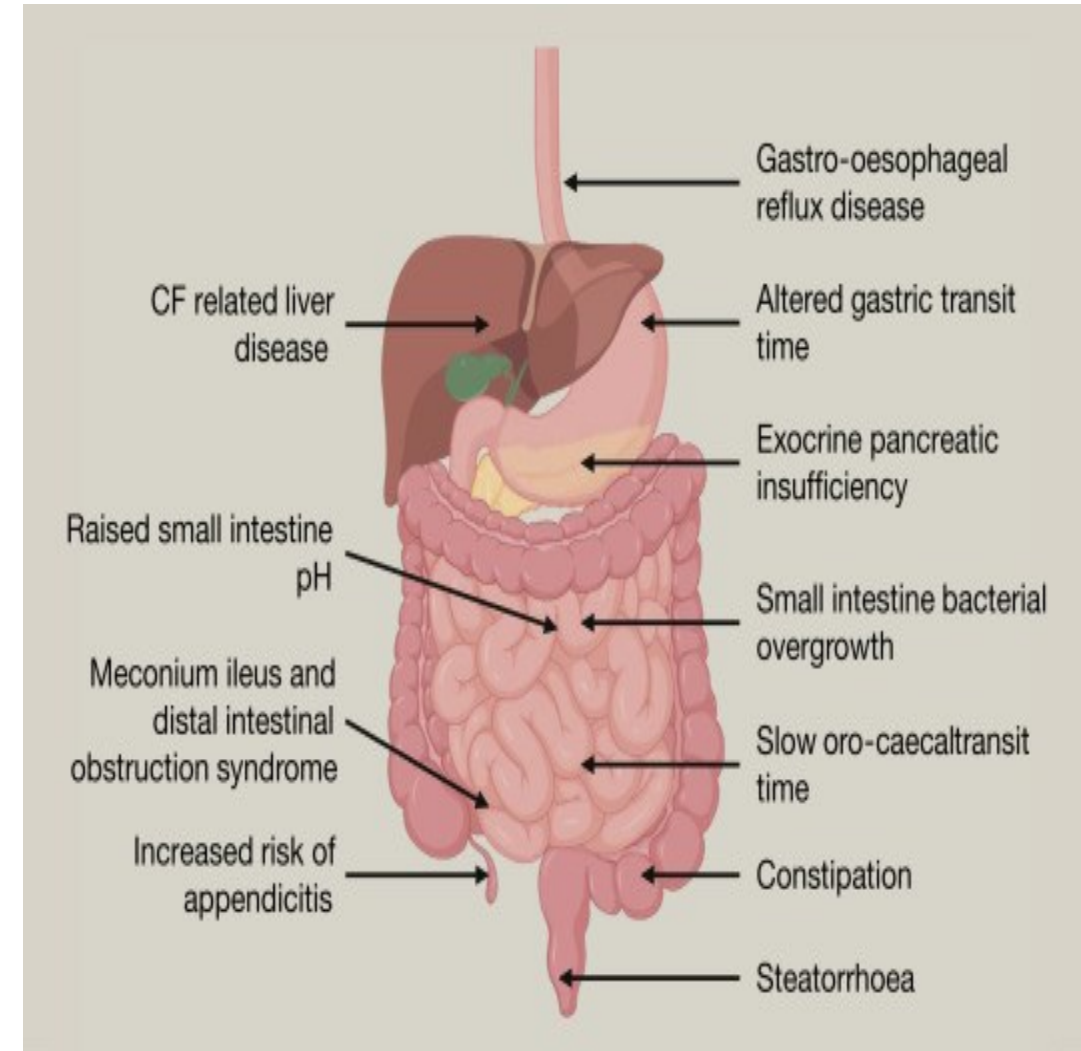


Cystic Fibrosis

Clinical Features:

Gastrointestinal symptoms of CF: common in children.

- Meconium ileus
- Failure to thrive
- Pancreatic disease
 - Pancreatitis
 - CF-related diabetes mellitus
 - Exocrine pancreatic insufficiency
 - Foul-smelling **steatorrhea**
 - Deficiency of fat-soluble vitamins
 - Malabsorption
 - Abdominal distention
 - Diarrhea
 - Hypoproteinemia

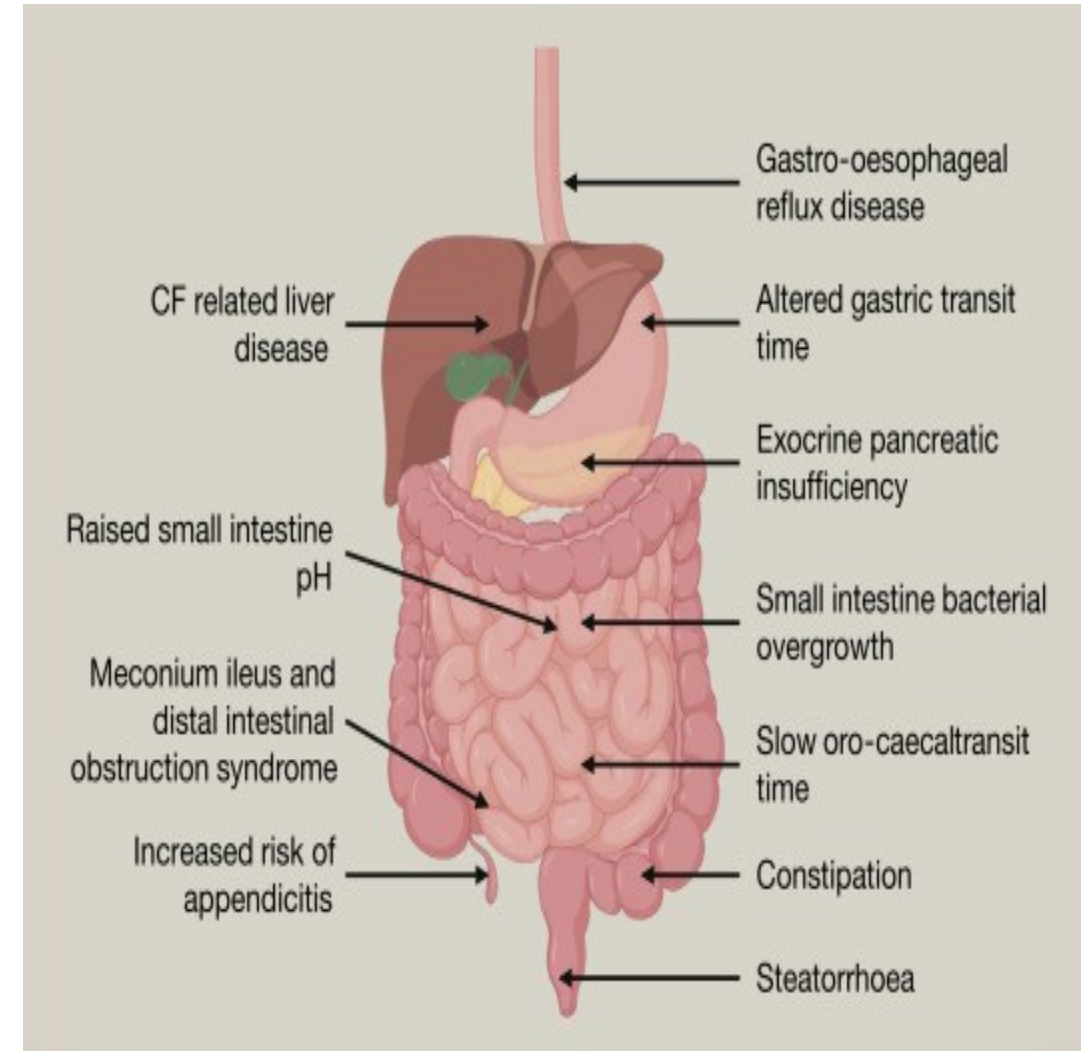


Cystic Fibrosis

Clinical Features:

Gastrointestinal symptoms of CF

- Liver and bile duct abnormalities
 - Cholecystolithiasis, cholestasis
 - Fatty metamorphosis of the liver, eventually progressing to liver cirrhosis
 - Biliary cirrhosis with portal hypertension.
- Intestinal obstruction: abdominal distention, pain, and a palpable mass
- Rectal prolapse



Cystic Fibrosis

Sweat glands

- Particularly **salty sweat**
- Possible electrolyte wasting

Musculoskeletal

- Frequent fractures due to osteopenia
- Kyphoscoliosis

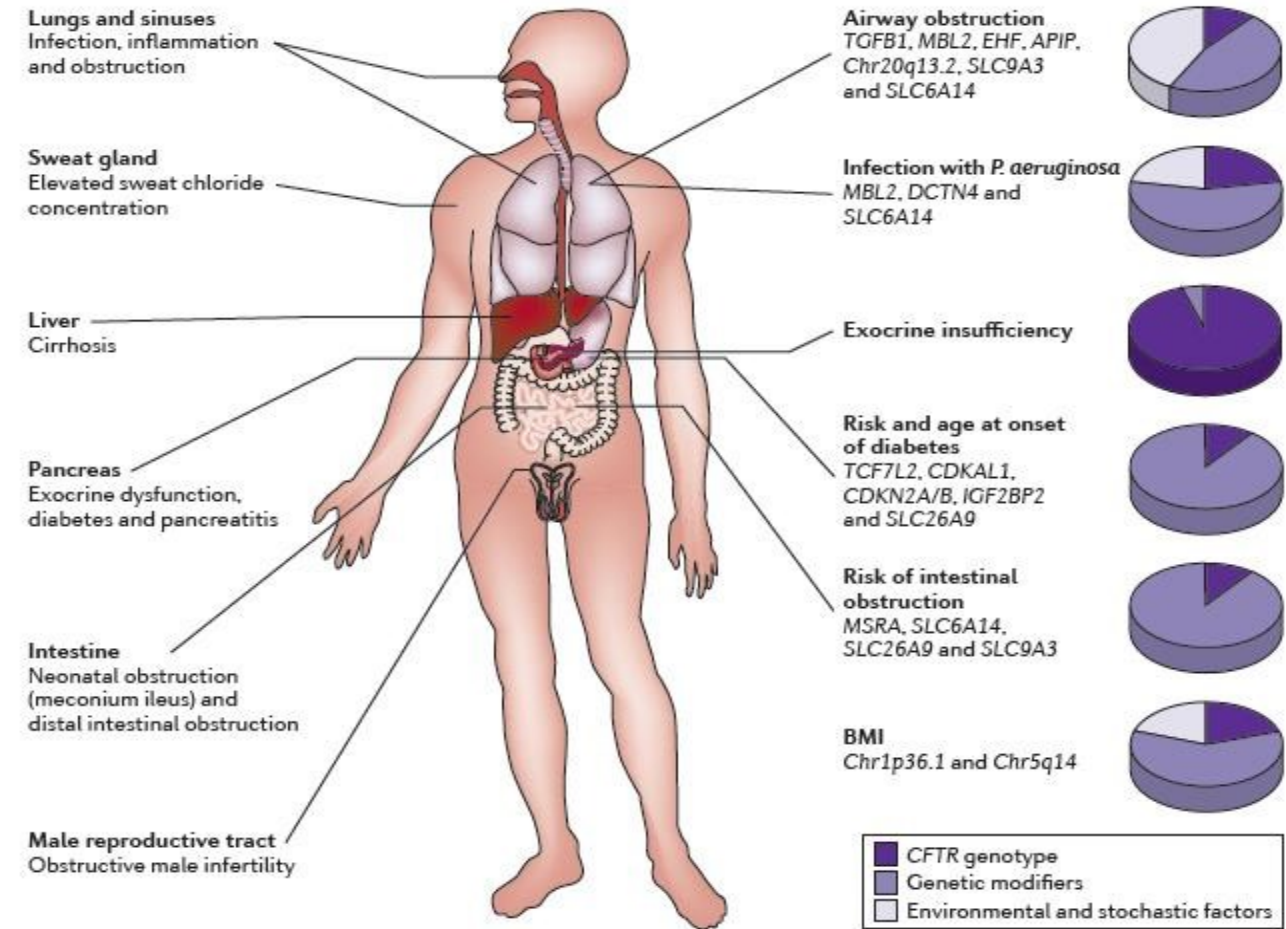
Urogenital

•Urinary

- Nephrolithiasis, nephrocalcinosis
- Frequent urinary tract infections

•Genital

- Men: usually infertile
 - **Obstructive** azoospermia is common, spermatogenesis may be intact
 - The vas deferens may be absent.
 - Undescended testicle
- Women: reduced fertility
 - Viscous cervical mucus can obstruct fertilization.
 - Menstrual abnormalities (e.g., amenorrhea)
- Delayed development of secondary sexual characters.



Cystic Fibrosis

Diagnostics:

- Sweat chloride test: with a chloride value ≥ 60 mmol/L
- Genotyping for CFTR mutations
- Nasal potential difference test: assesses Cl^- movement
- CXR
- PFTs: obstructive pattern



A



B

FIGURE 4.5. Cystic fibrosis. (A) Frontal CXR showing central cystic bronchiectasis (arrow) in a patient with CF. (B) Transaxial CT image showing cystic bronchiectasis (red arrow), with some bronchi containing impacted mucus (yellow arrow). (Reproduced with permission from USMLE-Rx.com.)

Cystic Fibrosis

Management:

Pharmacological interventions

- High-dose ibuprofen: can slow the progression of lung disease
Indication: Consider in patients 6–17 years of age with mild disease
- Bronchodilators: SABA, LABA
prior to inhaled therapies and/or chest physiotherapy , patients with episodes of airway hyperresponsiveness
Long-term use is not routinely recommended
- Mucolytics
Hypertonic saline nebulization (6–7% NaCl): mucociliary and osmotic effect that can improve mucociliary clearance and thin the mucus
Dornase alfa, aerosolized: a recombinant DNase that thins the mucus
- Antibiotics for infections (Pseudomonas!)
- Oxygen therapy:
Intermittent oxygen: desaturation during exercise and sleep
Long-term oxygen: hypoxia on room air and borderline hypoxia with evidence of cor pulmonale
- Double-lung transplant

Cystic Fibrosis

Nonpharmacological interventions

- Airway clearance techniques: a mainstay of CF treatment that loosens and mobilizes mucus secretions
- Vaccinations (influenza and pneumococcal)
- **Chest physiotherapy (CPT): postural drainage with percussion and/or clapping**
- Exercise (e.g., swimming, jogging, cycling)

Treat associated systems:

- **Pancreatic enzyme replacement**
- **Fat-soluble vitamin supplements**

To Summarize

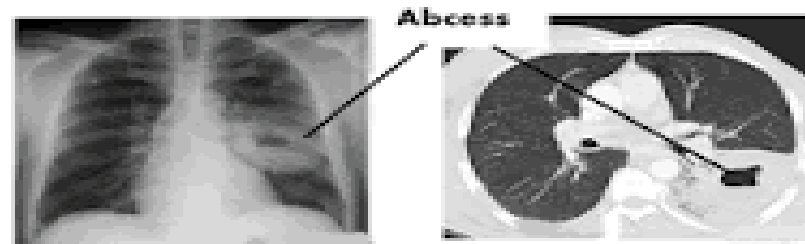
LUNG ABSCESS @rav7ks

Etiology:

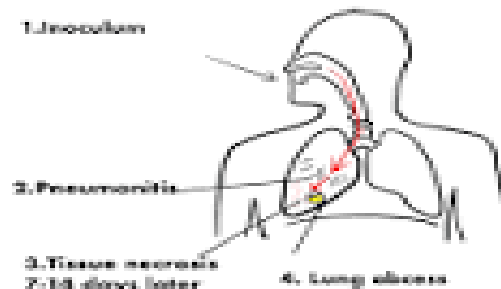
- necrosis of lung parenchyma by a polymicrobial infection
- most common cause: aspiration pneumonia
- DM risk factor for Klebsiella pneumonia
- Chronic alcoholism
- periodontal disease/ poor dentition/ gingival disease
- IV substance use
- Esophageal dysmotility
- Depressed consciousness
- Large volume feeding tubes

Clinical presentation:

- 80% have Fever 38 degrees
- chills, night sweats, fatigue ,unexplained weight loss
- Late disease:pleuritic chest pain, hemoptysis, dyspnea, productive cough
- putrid sputum/sour-tasting sputum



Pathophysiology:



Exams:

- poor dentition
- gingival crevice disease
- diminished breath sounds
- rales

Diagnosis:

- CBC: leukocytosis, anemia
- Sputum/ Blood cultures
- CXR (non- diagnostic in early disease)
- CXR upright or lat decubitus - air fluid level seen
- Chest CT

Management:

Antibiotics

- IV ampicillin-sulbactam
- Piperacillin/Tazobactam- pseudomonas
- carbapenems
- IV clindamycin in case of PCN allergy
- Switch to Oral antibiotics once stable

-10-15% require lobectomy or pneumonectomy if fail antibiotics

Consider surgical intervention:

- Fever > 10 days
- hemoptysis
- cavitation > 6 cm
- neoplasm, or hemorrhage

Treatment duration:

- 3 weeks or Continue antibiotic treatment until the chest radiograph shows a small, stable residual lesion or is clear. This generally requires several months of treatment
- **Delayed response:** evaluate for foreign body, cancer, or bronchial stenosis

Pathogens:

Aerobic Bacteria

Staph. Aureus.
 Legionella.
 H. Influenza.
 E. Coli.
 P. Aeruginosa.
 Strept. Pyogenes.
 K. Pneumonia.

Anaerobic bacteria

Bacteroides.
 Fusobacterium
 Peptostreptococcus
 Prevotella.

Fungal

Aspergillus.
 Blastomyces.
 dermatitis.
 Cryptococcus
 Coccidioides.

Parasite

Entamoeba Histolytica
 Paragonimus
 westermani

Other: TB

M. avium.
 M. Kansi

Differential diagnosis:

- vasculitis (eg. granulomatosis with polyangiitis).
- malignancy.
- aspirated foreign body.
- bronchiectasis

To Summarize

BRONCHIECTASIS

Bronchiectasis is a destructive lung disease characterized by chronic (permanent/irreversible) dilatation of the bronchi.

Symptoms:

General

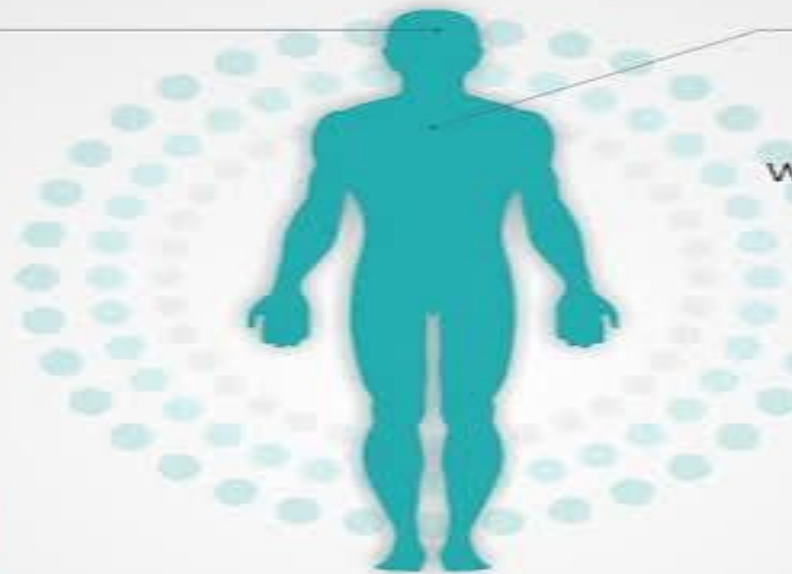


- Weight loss
- Poor Appetite
- Weakness
- Clubbing nails
- Drum Stick Nail



Lungs

- Paroxysmal Cough
- Cough purulent
- Dyspnoea
- Wheezing, Crepitation from lungs
- Blood in Cough
- Cough aggravated in morning
- Offensive cough by change of position



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CAUSES

- Cystic fibrosis
- Airway obstruction
- Lung Infections & injury
- Impaired host defences
- Autoimmune diseases
- Congenital
- Cigarette smoking

DIET & REGIMEN

- Avoidance of smoking
- Adequate hydration
- Adequate antioxidant
- Small, frequent feedings

TREATMENT

- Postural drainage
- Antibiotic therapy
- Treatment for co infection

MEDICINES

- Aconite
- Belladonna
- Bryonia alba
- Hepar Sulphur
- Kali Bichromicum
- Phosphorus

To Summarize

Cystic fibrosis

Epidemiology

Incidence: 1:3000–1:6000 in populations of northern European descent
Shortened life expectancy (~45 years)

Etiology

Autosomal recessive disorder
Mutation (most commonly a $\Delta F508$ Phenylalanine deletion) in the CFTR gene on the long arm of chromosome 7 that leads to chloride channel defects in exocrine glands

Diagnostics

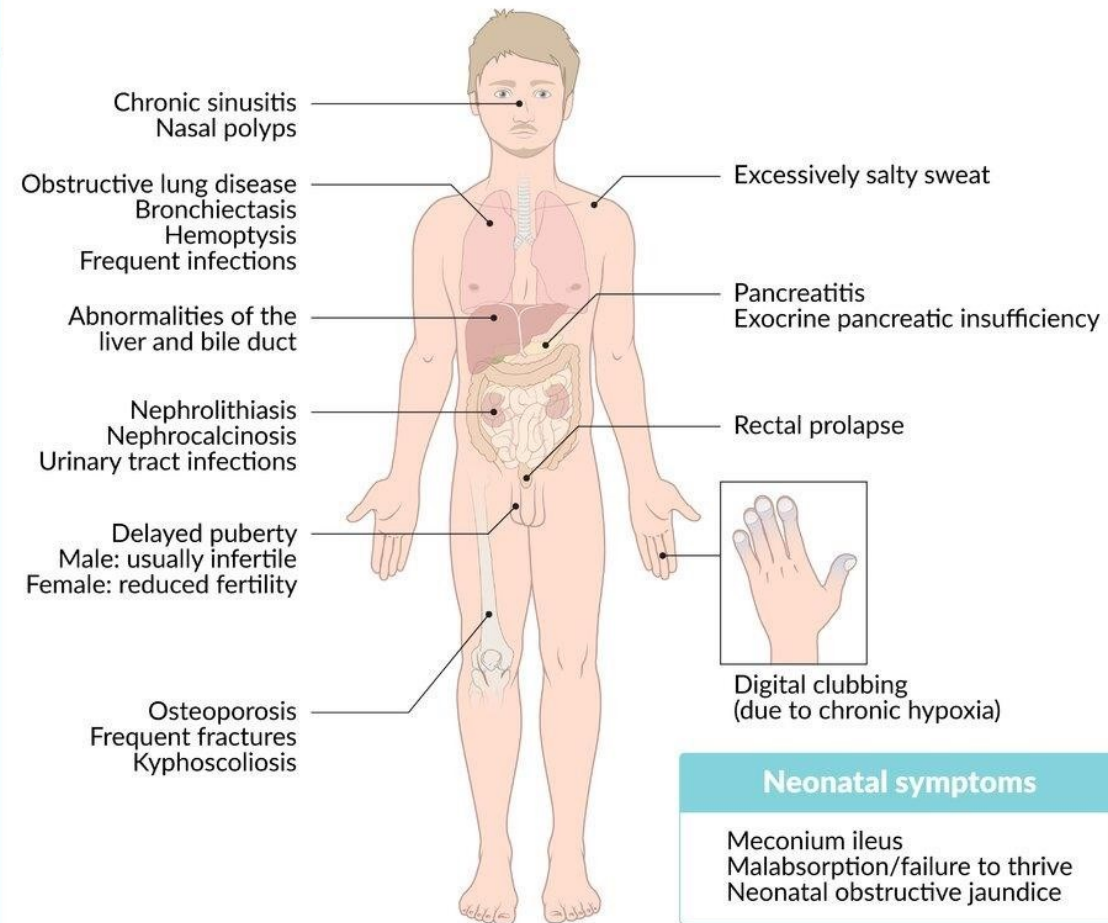
Clinical suspicion for CF (positive newborn screen, family history, signs and/or symptoms) AND laboratory evidence of CFTR dysfunction (sweat chloride testing, genetic analysis, physiologic testing)

Management

CFTR modulator therapy
Airway clearance and pharmacological preservation of lung function
Antibiotics for chronic lung infections and acute exacerbations
Optimization of nutrition (e.g., pancreatic enzyme supplements)
Lung transplant

Complications

Progressive respiratory decline and, ultimately, respiratory failure
Opportunistic lung infections (e.g., *Pseudomonas aeruginosa*)
CF-related diseases: liver disease (CFLD) and diabetes (CFRD)
Gastrointestinal obstruction (e.g., SBO, DIOS)



Case Scenario 1

A 65-year-old man is brought to the emergency department after coughing up copious amounts of blood-tinged sputum at his nursing home. He has recently had an upper respiratory tract infection that was treated with antibiotics. He has a long-standing history of productive cough that has worsened since he had a stroke 3 years ago. He smoked a pack of cigarettes daily for 40 years until the stroke, after which he quit. The patient appears distressed and short of breath. His temperature is 38°C (100.4°F), pulse is 92/min, and blood pressure is 145/85 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 92%. Physical examination shows digital clubbing and cyanosis of the lips. Coarse crackles are heard throughout the lung fields. An x-ray of the chest shows increased translucency and tram-track opacities in the right lower lung field. Which of the following is the most likely diagnosis?

- a. Bronchiectasis
- b. Chronic bronchitis
- c. Aspiration pneumonia
- d. Emphysema
- e. Pulmonary embolism
- f. Lung cancer

Case Scenario 2

A 2-year-old boy is brought to the physician because of an increasing productive cough with a moderate amount of white phlegm for the past week. He has been treated for pneumonia with antibiotic therapy four times over the past year. A chest x-ray performed 3 months ago showed no anatomical abnormalities. He has had multiple episodes of foul-smelling, bulky, greasy stools. He is at the 3rd percentile for height and at the 5th percentile for weight. His temperature is 38.0°C (100.4°F), pulse is 132/min, and respirations are 44/min. A few inspiratory crackles are heard throughout the lung fields. The abdomen is soft and nontender. The remainder of the examination shows no abnormalities. Which of the following is the best initial test to determine the underlying etiology of this patient's illness?

- a. CT scan of the chest
- b. X-ray of the chest
- c. Pulmonary function tests
- d. Serum immunoglobulin level
- e. Sweat chloride test
- f. Stool analysis
- g. DNA phenotyping

Questions?



References & further reading

- Step-up to medicine 5th edition
- Harrison's Principles of Internal Medicine 20th edition
- Amboss
- Cecil Essentials of Medicine 10th edition
- First Aid for the internal medicine board 4th edition

Thank You